The landslide susceptibility values are derived directly from the logistic regression. The landslide frequency and affected area values are obtained from the combination of susceptibility values and landslide frequency and size (see Step 3 for details).

From a visual interpretation of the two susceptibility classes with the highest values, the Bishenyi sub-catchment displays the susceptibility pattern that is the most evenly distributed across the landscape, whereas Rwandex-Magerwa sub-catchment is the landscape where concentration of the highest susceptibility areas is the most concentrated (Figure 5).

The distribution of the deep-seated landslides is different from that of the shallow landslide susceptibility. The Kamembe-Gihundwe sub-catchment is affected by 15 old landslides while no such slope failure processes are observed in the Rwandex-Magerwa landscape (Figure 5).The largest old landslides is located in the Bishenyi. This is also in that sub-catchment that the five recent deep-seated landslides are present (Figure 5).

The difference between the susceptibility patterns of shallow and deep-seated landslides is not something uncommon, and particularly at this scale of investigation. As explained in the inventory section (Step 1), deep-seated landslides can have a complex origin that spans a very long period of changing environmental conditions while shallow landslides reflect more the current environment where human-activities (land use/cover and roads) also play a role.

At the local scale, the exact location of deep-seated landslides highly depends on heterogeneous local conditions associated with the lithology and hydrogeology (e.g. Dille et al., 20019) that are much more difficult to constrain than slope and other topographic conditions that usually have a stronger direct control on shallow slope failures (Sidle and Bogaard, 2016).